

CS 457 / 657

Final Project

Project Goal

The goal of the final project is for you to demonstrate your ability to design, implement, and integrate a database into a simple software system.

Project Requirements

System Requirements (system = software + database)

1. There must be some information going into the system which persists in the database.
2. There must be some information coming out of the system which is generated by the database.
3. There must be a UI (which can fulfill the requirements listed above).
 - a. There is no requirement around how the UI should be implemented; GUIs, web apps, CLIs, etc. are all acceptable.

Software Requirements

1. Any code that is not SQL should be written in Python3.
 - a. You must use either a Python3 library or framework to interact with the database.
2. Any code that is not SQL should use an Object-Oriented approach.
3. Interactions between the database and outside of the system must pass through the software layer.
 - a. This requirement applies to while the system is running. If your outside data source is used to “seed” the database with information, that import can happen outside of the software layer.

Database Requirements

1. The database management system must be PostgreSQL.
2. There must be at least one “outside” data source (not generated within the software system).
 - a. The software system can have functionality to save data from the “outside” data source to the database.
3. There must be at least 5 tables in the database's schema which are involved in the system's processes.
 - a. The system must at least be complex enough to meaningfully incorporate 5 or more tables into achieving its overall goal.
4. Each table should be in 3NF (third normal form).
5. If a supplemental key is used, there must be an additional uniqueness constraint which enforces the actual primary key.
 - a. Natural keys are acceptable as primary keys.
 - b. Composite keys are acceptable as primary keys.